

applies for power semiconductors, silicon carbide, and for gallium nitride products.

The sector that we are focusing on right now is medium-to-high-power, energy sector, renewables, automotive, defence, and aerospace. We will be selling products under the Visicon brand in India and under Silicon Power brand globally.

Can you explain the three stages and the three phases elaborately?

The PLI scheme is one application that includes all phases—phase 1, phase 2, and phase 3. The three phases address the critical components of the supply chain of the silicon carbide ecosystem. These start with the initial material phase, from processing to packaging under Make in India, and then working towards the end use of these products.

It begins with material, product packaging, and finally fabrication.

Will you be providing these services to other silicon original equipment manufacturers (OEMs) as well?

In India, there is no company that manufactures silicon carbide. We will definitely look at all the possibilities while defining economies of scale by working with other OEMs. First it would be our company from Oregon, then there may be others who are looking for epi services or other services that we offer, or plan to offer.

At this moment, there is a global shortage of epi-material. Before we start to manufacture, we would provide that service to global players so that we can start generating revenue.

What is the advantage that you have figured out while doing this in India, while a lot of others want to stay away?

The secret sauce is the fact that we want to learn to walk first before we start to run. That is what we have done all along. My philosophy is why not!

When in 2000 I wanted to transfer technology from USA to India, the biggest challenge I faced was the cost competition due to the presence of Chinese companies in the market. I had two choices to serve the same customers—either I buy components from China, or I find another high-quality lower-price manufacturing centre. I chose the latter.

We have always tried to set-up a business model where the break-even is not eight to ten years but more like two to three years.

How did you handle the Chinese competition in Ruttonsha?

We started by focusing on the quality of the products. I experienced that myself when we had ordered some components from China. Initially when we asked for samples, we got excellent products. But when the bulk of 950 components came, our customers returned 900 of those. This is the problem that we addressed at Ruttonsha.

There is a team of 150 people at Ruttonsha that makes sure that every component is tested before it is shipped to the customer. If we are successful in the plan we have made, we will employ around 80 to 100 more people in the same facility in the next two to three years.

In fact, we have recently inaugurated our facility at Halol and the ceremony was attended by people from different IITs in India. We are trying to coordinate with an integrated resource plan so that we can take care of the things that we need, as we need them.

When you say resource, what all are you referring to?

By resources I mean manpower—students, graduates, and post-docs. More importantly, the government of India has invested heavily in providing state-of-the-art equipment to a lot of universities. We also want to make sure that their facility is fully utilised. We can help them in training the students and we can use that equipment on some kind of service basis.

We have already signed contracts with IIT Delhi's (IIT-D) Rajendra Singh. He is working with us to help develop processes in terms of reliability and others. There is equipment at IIT Delhi that we can use for research and development and, in turn, train the students on a practical basis.

The professors who came to the inauguration ceremony complimented us saying that their students can now practically see how the manufacturing takes place. Emroj Hussain, who is one of our first epitaxial engineers at the Halol plant, got his PhD from Tata Institute of Fundamental Research (TIFR) and was confused whether to stay in India or take up a job in the USA. He was very happy when Visicon offered him a job.

Do you think India has the manpower for such projects?

I think India has the resources needed. When there are so many Indians working outside India in this field, I think resources are available.

Now, no matter how big of a university you come out of, there is always an aspect of practical training. What you learn in books and how you implement the same is a little bit different. Being able to work with IITs, I think, is a step forward in teaching those students how to work with the needs of the industry.

Any immediate hiring happening at your firm?

We expect to add around 200 people in the next three years.

Is there an opportunity for more universities to collaborate with you? What can they provide and expect in return?

In any kind of a research collaboration, technical ability of both the partners is always the number-one aspect. What we need to do is find core expertise of a university and then work collaboratively on a research project. We do not want to do the same things at different places.

Our aim is to work with centres of excellence and guide them in the best possible pragmatic way. We are open to talks with universities and academic institutes, and collaborations in any way we can.

At what stage did your team start interacting with academia to support your research & development initiative? Was this being done at your earlier firm itself, or was a different connection established for the new firm?

I believe in efficiency and leverage working together as everyone achieves more as a team. I have always followed this model since my first job at Electric Power Research Institute (EPRI) in the eighties. Working with academia helps us optimally explore state-of-the-art solutions coupled with my pragmatic solution-driven approach. We aggressively look to create manufacturable, usable, sellable technology (MUST). I also believe and follow the integrated globalisation model.

We did initiate our interaction with IIT Delhi, even before our Visicon plan was put on paper. Besides IIT Delhi, we are also engaged with both IIT Bombay (IIT B) and TIFR and shall be leveraging some of their expertise with SiC manufacturing, which is a new technology.

In the US, I also work with the best and the brightest of minds at universities and am honoured to have known and worked with Prof. Agrawal at